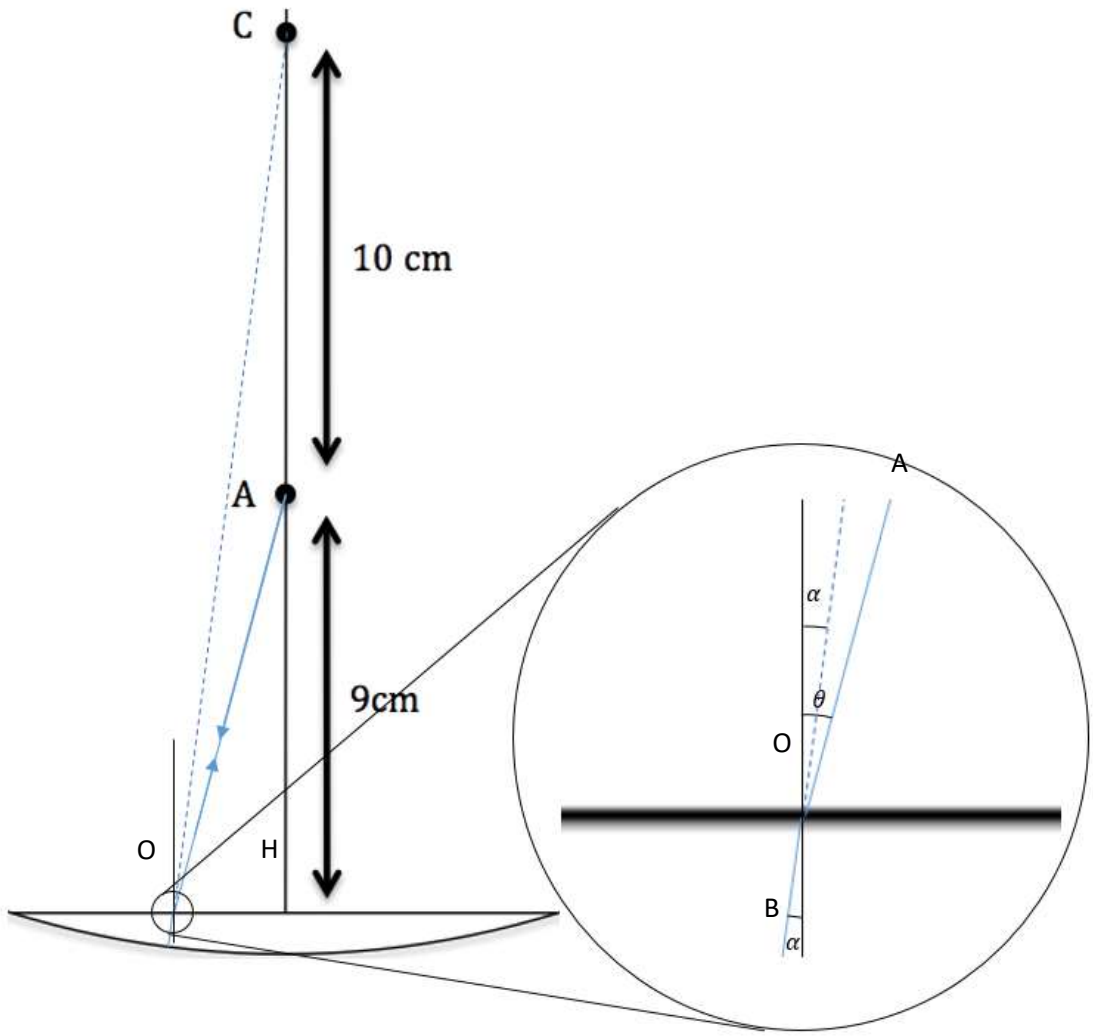
Question 1	a	b	c	d	e	f
Question 2	a	b	c	d	e	f
Question 3	a	b	c	d	e	f
Question 4	a	b	c	d	e	f
Question 5	a	b	c	d	e	f
Question 6	a	b	c	d	e	f
Question 7	a	b	c	d	e	f
Question 8	a	b	c	d	e	f
Question 9	a	b	c	d	e	f
Question 10	a	b	c	d	e	f
Question 11	a	b	c	d	e	f
Question 12	a	b	c	d	e	f
Question 13	a	b	c	d	e	f
Question 14	a	b	c	d	e	f
Question 15	a	b	c	d	e	f
Question 16	a	b	c	d	e	f
Question 17	a	b	c	d	e	f
Question 18	a	b	c	d	e	f
Question 19	a	b	c	d	e	f
Question 20	a	b	c	d	e	f
Question 21	a	b	c	d	e	f
Question 22	a	b	c	d	e	f
Question 23	a	b	c	d	e	f
Question 24	a	b	c	d	e	f
Question 25	a	b	c	d	e	f

Solution to Problem 1



To have the image of A fall on A, all incoming light arrays from A should get reflected back to their initial path. Therefore, the line OB should pass from the center (C).

$$\tan\alpha = \frac{OH}{19cm} \approx \alpha$$

$$n\sin\alpha = \sin\ \theta$$

For small angles:

$$n\,\alpha \approx n\frac{OH}{19cm} \approx\ \tan\theta = \frac{OH}{9cm}$$

Therefore:

$$n = \frac{19}{9} \approx 2.11$$

Solution to Problem 2

From the photos one can measure that the height of the woman of 1.68 m corresponded to 35 mm so the conversion was 0.048 m/mm.

One could measure the change of height of center of gravity of the person from the position in the air, at the top of the jump to the bottom when the spring was fully compressed. It was 7 mm which corresponds to 0.34 m. One could also measure the compression of the spring x (how much shorter the bottom of the pogo stick rod looks) to be 5 mm or 0.24 m.

All gravitational potential energy is converted to the elastic potential energy of the spring

$mgh = kx^2/2$

where m = 60 kg (mass of the pogo stick can be neglected or, as some students dis estimated to be of the order of 5kg), h = 0.34 m, x = 0.24 m, k is the elastic constant of the spring.

$k = 2mgh/x^2 = 6900$ N/m

When in contact with ground the system can be considered to be an harmonic oscillator (mass on a spring) with characteristic frequency of $\omega = \sqrt{k/m} = 11$rad/s

Period of oscillations is $T = 2\pi/\omega = 0.58$ s

The equilibrium position (when the woman just stands on the stick is at defined by equilibrium of forces $mg = kx_e$ which give $x_e = mg/k = 0.085$ m below the position, when the spring is fully extended.

The time when the system is in contact with ground can be well approximated as half of the period or calculated exactly as the time of the oscillation between the positions of 0.24 m above the minimum.

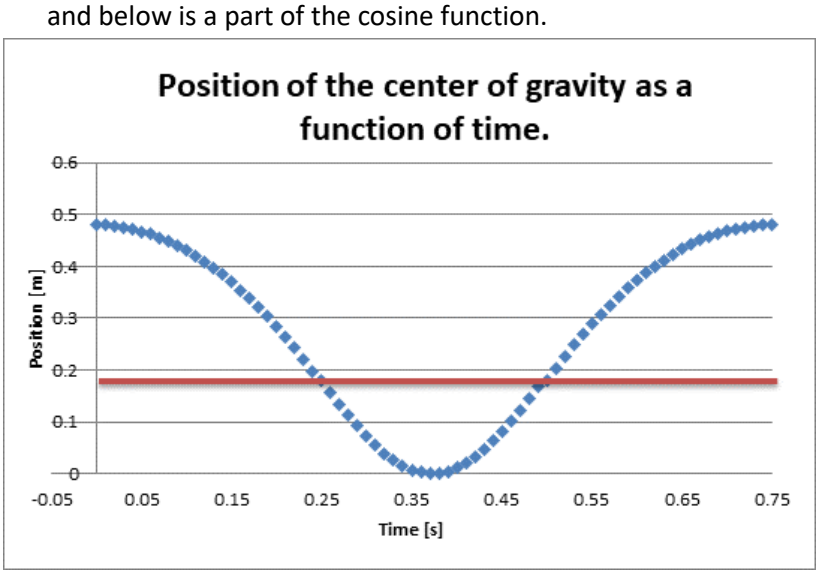
The time in air T_a is twice the time it takes to drop from 0.34 m to 0.155 m (0.24 – 0.085) if the approximation was used or from 0.48 m to 0.24 m for exact calculation.

$T_a = 2\sqrt{2h2}/g = 0.39$ s

Total period is $T_a + ½ T = 0.68$ s

One expects at least 10% uncertainty in these calculations.

The graph of the position of the center of gravity as a function of time is shown below. The zero of the position scale corresponds the lowest point. The function above 0.18 m (red line) are parts of a parabola and below is a part of the cosine function.



Solution to Problem 3

June 1, 2017

3a) Consider a slice through the centre of the water bubble. This cut has perimeter $l = 2\pi r$ and area $A = \pi r^2$. The pressure in the water bubble can be calculated by considering the force exerted by the bubble on this slice and diving by the area of the slice,

$$P = \frac{F}{A} = \frac{F}{\pi r^2}. \tag{1}$$

The force is given by $F = \sigma l$, where l is in this case the perimeter of the slice. This leads to a pressure,

$$P = \frac{\sigma 2\pi r}{\pi r^2} = \frac{2\sigma}{r}. \tag{2}$$

We now substitute in the radius $r = 0.05\,\mathrm{m}$ and the surface tension constant $\sigma_{water} = 0.073\,\mathrm{N/m}$. This is the pressure exerted by the bubble, so is the difference in pressure between inside and outside the bubble.

3b) Note that since $P \propto 1/r$ from above, the smaller bubble will have higher pressure, so the water will flow from the smaller to the larger bubble.
The pressure difference is given by

$$\delta p = 2\sigma \left(\frac{1}{r_1} - \frac{1}{r_2} \right). \tag{3}$$

We set this equal to the pressure difference across a pipe, as given by the formula in the problem,

$$2\sigma \left(\frac{1}{r_1} - \frac{1}{r_2} \right) = 32\mu L V_{av}/D^2. \tag{4}$$

Next we solve for V_{av}, the average velocity in the pipe,

$$V_{av} = \frac{2\sigma D^2}{32\mu L} \left(\frac{1}{r_1} - \frac{1}{r_2} \right). \tag{5}$$

Substituting in $D = 1\,\mathrm{mm}$, $r_1 = 2\,\mathrm{cm}$, $r_2 = 5\,\mathrm{cm}$, and $\mu_{water} = 0.001002\ \mathrm{N} \cdot \mathrm{s/m^2}$ we determine that the initial velocity of water in the pipe is $V_{av} = 0.0027\,\mathrm{m/s}$.

c) A good estimate is given by calculating the average velocity twice, once at the initial radii (as was already done in part b) and once when the small ball has a radius roughly given by the radius of the straw. Averaging these two velocities one finds some effective velocity V. We can then take $V \times$ Area of the straw as an effective flow rate F, so that V/F for V the initial volume of the small sphere gives the estimated time. Using this method ones finds that it will take about 14 minutes.

A more naive estimate is to use the initial flow rate, in this case the ball disappears after 264 minutes. Another possibility is to use the velocity when the ball is half gone, this gives 88 minutes.